



WannaCry Ransomware - Static Analysis Report

Analysis Date: January 15, 2026

Samples Analyzed: 2 (Dropper + Payload)

Executive Summary

This report presents a comprehensive static analysis of the **WannaCry ransomware**, one of the most significant cyber attacks in history. Two components were analyzed:

1. **Dropper Component** - Responsible for payload extraction and deployment
2. **Payload Component** - Contains the encryption logic and kill switch mechanism

Key Findings:

- ☒ Confirmed WannaCry ransomware family
- ☒ Kill switch domain identified and analyzed
- ☒ Dropper and payload mechanisms fully documented
- ☒ Multiple IOCs extracted for detection

Verdict: **HIGHLY DANGEROUS** - Ransomware with worm capabilities

1. Dropper Component Analysis

1.1 Overview

The dropper is a self-extracting executable that deploys the main ransomware payload to the victim system.

Entry Point: 004077ba

1.2 Execution Flow

```
graph TD
    A[Entry Point] --> B[Check Command Line Args]
    B --> C{Args == 2?}
    C -->|Yes| D[Copy to tasksche.exe]
    C -->|No| E[Set Working Directory]
```

```
D → E
E → F[Extract Resource 0x80a]
F → G[Unzip with Password]
G → H[Set Permissions]
H → I[Load t.wnry DLL]
I → J[Call TaskStart Export]
J → K[Execute Ransomware]
```

1.3 Key Behaviors

Persistence Mechanism

```
# Copies itself to system directory
tasksche.exe
```

```
# Hides files
attrib +h .
```

```
# Grants full permissions
icacls . /grant Everyone:F /T /C /Q
```

Payload Extraction

- **Resource ID:** 0x80a
- **Compression:** ZIP (zlib 1.1.3 / unzip 0.15)
- **Password:** WNCry@2017
- **Output:** t.wnry (Main payload DLL)

Dynamic Loading

Uses custom `GetProcAddress` implementation at 00402924 to resolve the `TaskStart` export from `t.wnry`, avoiding standard API hooks.

1.4 Dropper IOCs

Type	Value	Description
Filename	tasksche.exe	Persistence copy
Password	WNCry@2017	ZIP extraction password
DLL	t.wnry	Main payload module
Export	TaskStart	Payload entry point
Config	c.wnry	Configuration file

Type	Value	Description
Mutex	Global\\MsWinZonesCacheCounterMutexA	Synchronization mutex

2. Kill Switch Analysis

2.1 Discovery

[!IMPORTANT]

Kill Switch Domain: <http://www.iuqerfsodp9ifjaposdfjhgosurijfaewrrergwea.com>

Location: Address `004313d0` in payload component

Function: `KillSwitch_Check` at `00408140`

2.2 Implementation

The kill switch is checked at the very beginning of the payload execution:

```
undefined4 KillSwitch_Check(void)
{
    undefined4 hInternet;
    int hUrl;
    char killswitch_url[57];

    // Copy kill switch URL
    strcpy(killswitch_url, "<http://www.iuqerfsodp9ifjaposdfjhgosurijfaewrrergwea.co
m>");

    // Initialize WinINet
    hInternet = InternetOpenA(NULL, INTERNET_OPEN_TYPE_DIRECT);

    // Attempt to connect to kill switch domain
    hUrl = InternetOpenUrlA(hInternet, killswitch_url, NULL, 0, 0x84000000, 0);

    if (hUrl == 0) {
        // Domain NOT reachable → PROCEED with ransomware
        InternetCloseHandle(hInternet);
        InternetCloseHandle(0);
        ContinueExecution(); // Encrypt files
        return 0;
    }

    // Domain IS reachable → STOP ransomware
    InternetCloseHandle(hInternet);
    InternetCloseHandle(hUrl);
}
```

```
return 0; // Exit gracefully
}
```

2.3 Kill Switch Logic

```
graph LR
    A[Start Execution] --> B[Call KillSwitch_Check]
    B --> C{Try Connect to Domain}
    C -->|Success| D[Domain Reachable]
    C -->|Failure| E[Domain Unreachable]
    D --> F[EXIT - No Encryption]
    E --> G[PROCEED - Encrypt Files]

    style F fill:#90EE90
    style G fill:#FFB6C6
```

2.4 Historical Context

Date: May 12, 2017

Researcher: Marcus Hutchins (@MalwareTechBlog)

Marcus Hutchins discovered this unregistered domain in the malware code and registered it for \$10.69, inadvertently activating the kill switch and halting the global WannaCry outbreak. By making the domain reachable, all WannaCry instances worldwide began exiting before encrypting files.

2.5 Why the Kill Switch Exists

Theories:

1. **Sandbox Evasion** - Malware analysts often use sandboxes with fake internet connections. If all domains resolve, the malware assumes it's in a sandbox and exits.
2. **Developer Backdoor** - Intentional emergency stop mechanism.
3. **Testing Artifact** - Left over from development/testing phase.

3. Ransomware Capabilities

3.1 Targeted File Extensions

WannaCry targets **200+ file types**, including:

Documents: .doc , .docx , .pdf , .xls , .xlsx , .ppt , .pptx , .odt , .txt

Databases: .sql , .mdb , .accdb , .dbf , .sqlite3 , .sqlitedb

Media: .jpg , .jpeg , .png , .gif , .bmp , .mp3 , .mp4 , .avi , .mov

Development: .cpp , .java , .php , .asp , .jsp , .py , .rb , .js

Backups & Archives: .zip , .rar , .tar , .bak , .backup

Certificates & Keys: .der , .pfx , .key , .crt , .pem , .p12

3.2 Encryption Method

- **Algorithm:** AES + RSA (via CryptoAPI)
- **Provider:** Microsoft Enhanced RSA and AES Cryptographic Provider
- **APIs Used:** CryptAcquireContextA , CryptGenKey , CryptEncrypt , CryptImportKey

3.3 Ransom Payment

Bitcoin Addresses:

13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94
12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw
115p7UMMngo1pMvvpHjRdfJNXj6LrLn

Ransom Note: @WanaDecryptor@.exe displays ransom demands

4. Network Indicators

4.1 Kill Switch Domain

Indicator	Value
Domain	www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com
Full URL	http://www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com
Protocol	HTTP (Port 80)
Purpose	Kill switch check

4.2 Network Behavior

- **WinINet APIs:** InternetOpenA , InternetOpenUrlA , InternetCloseHandle
- **Connection Type:** Direct (no proxy)
- **Flags:** 0x84000000 (INTERNET_FLAG_NO_CACHE_WRITE | INTERNET_FLAG_RELOAD)

5. Complete IOC Summary

5.1 File Indicators

Type	Value
Dropper	Various names (original sample)
Persistence	tasksche.exe
Payload DLL	t.wnry
Config	c.wnry
Signature	WANACRY! string at 0040eb7c

5.2 Network Indicators

Type	Value
Kill Switch	www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergrwea.com
Bitcoin 1	13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94
Bitcoin 2	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw
Bitcoin 3	115p7UMMngo1pMvvpHijcRdfJNXj6LrLn

5.3 Registry/System Indicators

Type	Value
Mutex	Global\\MsWinZonesCacheCounterMutexA
Commands	attrib +h .
Commands	icaccls . /grant Everyone:F /T /C /Q
Service	Creates service for persistence

6. Detection & Mitigation

6.1 YARA Rule

```
rule WannaCry_Ransomware {
  meta:
    description = "Detects WannaCry ransomware"
    date = "2026-01-15"

  strings:
    $killswitch = "iuqerfsodp9ifjaposdfjhgosurijfaewrwergrwea.com"
    $password = "WNcry@2o17"
    $payload = "t.wnry"
    $signature = "WANACRY!"
    $bitcoin1 = "13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94"
    $mutex = "Global\\MsWinZonesCacheCounterMutexA"
```

```
condition:
  3 of them
}
```

6.2 Network Detection

Snort/Suricata Rule:

```
alert http any any → any any (msg:"WannaCry Kill Switch Domain";
  content:"iuqerfsodp9ifjaposdfjhgosurijfaewrwergrgwea.com";
  sid:1000001; rev:1;)
```

6.3 Mitigation Strategies

1. Kill Switch Activation

- Ensure kill switch domain remains registered and reachable
- Monitor for connection attempts to the domain

2. Patch Management

- Apply MS17-010 (EternalBlue) patch
- Keep systems updated

3. Network Segmentation

- Limit SMB (port 445) exposure
- Implement network segmentation

4. Backup Strategy

- Maintain offline backups
- Test backup restoration regularly

5. Endpoint Protection

- Deploy behavioral detection
- Monitor for suspicious file encryption activity

7. Technical Deep Dive

7.1 Custom GetProcAddress Implementation

WannaCry implements its own export resolution at `00402924` to avoid API hooks:

```

int CustomGetProcAddress(int *pe_base, char *export_name)
{
    int export_dir_rva = *(int *)(*pe_base + 0x7c);
    if (export_dir_rva != 0) {
        int export_dir = export_dir_rva + pe_base[1];
        int num_names = *(int *)(export_dir + 0x18);

        // Walk export name table
        int *name_table = (int *)((*int *) (export_dir + 0x20) + pe_base[1]);
        ushort *ordinal_table = (ushort *)((*int *) (export_dir + 0x24) + pe_base[1]);

        for (int i = 0; i < num_names; i++) {
            if (_stricmp(export_name, (char *) (name_table[i] + pe_base[1])) == 0) {
                int ordinal = ordinal_table[i];
                int *addr_table = (int *)((*int *) (export_dir + 0x1c) + pe_base[1]);
                return addr_table[ordinal] + pe_base[1];
            }
        }
    }
    return 0;
}

```

7.2 Resource Extraction Process

1. **Find Resource:** `FindResourceA(hModule, 0x80a, type)`
2. **Load Resource:** `LoadResource(hModule, hResInfo)`
3. **Lock Resource:** `LockResource(hResData)` → Get pointer to compressed data
4. **Decompress:** Custom zlib implementation decompresses ZIP archive
5. **Extract Files:** Unzip using password `WNCry@2ol7`
6. **Deploy:** Write files to disk and set permissions

8. Conclusion

WannaCry represents a sophisticated ransomware with worm capabilities. The analysis revealed:

Strengths:

- Multi-stage deployment (dropper + payload)
- Custom API resolution to evade hooks
- Encrypted payload storage

- Worm propagation via EternalBlue exploit

Weaknesses:

- Hardcoded kill switch domain (single point of failure)
- No domain generation algorithm (DGA)
- Unencrypted strings in payload
- Predictable file naming

Impact:

The May 2017 outbreak affected over 200,000 computers across 150 countries, causing billions in damages. The accidental discovery and activation of the kill switch prevented significantly more damage.

Recommendations:

Organizations should maintain updated systems, implement network segmentation, monitor for kill switch domain connections, and maintain robust offline backup strategies.

9. References

- **CVE-2017-0144** (EternalBlue SMB vulnerability)
 - **MS17-010** (Security patch)
 - **MITRE ATT&CK:** T1486 (Data Encrypted for Impact)
 - **Kill Switch Registration:** May 12, 2017 by Marcus Hutchins
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Report Classification: Technical Analysis

Last Updated: January 15, 2026